Curiosity and Affect-Driven Cognitive Architecture for HRI

Letícia Berto, Ana Tanevska, Azamor Cirne, Paula Costa, Alexandre Simões, Ricardo Gudwin, Francesco Rea, Esther Colombini, and Alessandra Sciutti

***Abstract*—This study explores how humans and cognitive robots with different value systems and motivations understand each other's needs in free-form interactions. We developed a cognitive architecture that links sensing and perception to internal motivation and an intrinsic value system for determining actions. Inspired by young children's needs, this architecture includes three drives: learning, interaction, and recharging, each with varying dependence on the human partner. We aimed to assess how experimentally changing the importance of these drives within a fixed architecture affects interaction dynamics with human partners (acting as caregivers) and their understanding of the robot's needs. By adjusting the learning and interaction drives, we created two robot profiles: Playful, which prioritizes environmental exploration and playfulness to reduce boredom, and Social, which focuses on social interaction through touch and visual contact to increase comfort. Our findings show that changing the importance of these drives produces distinct behaviors and human perceptions. Robot behaviors matched their profiles, and participants adapted their responses accordingly. Participants identified and attributed distinct traits to each robot without knowing the specific profiles. Despite variability among human partners, the robots, especially the playful one, were generally well understood by most participants.**

***Index Terms*—Human-robot interaction, intrinsic motivation, cognitive architecture, robot profiles, affective computing.**

Received 30 June 2024; revised 3 February 2025; accepted 7 March 2025. Date of publication 14 March 2025; date of current version 25 May 2026. This work was supported in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001, in part by São Paulo Research Foundation (FAPESP), Brasil under Grant 2021/07050-0, in part by the Ministry of Science, Technology, and Innovation of Brazil, with resources granted by the Federal Law 8.248 of October 23, 1991, under the PPI-Softex. The project was coordinated by Softex and published as Intelligent agents for mobile platforms based on Cognitive Architecture technology [01245.003479/2024-10]; in part by the European Research Council's Horizon 2020 programme, under Grant 804388, in part by wHiSPER, and in part by NextGenerationEU PNRR MUR - M4C2 project under Grant PE00000013 - FAIR. Recommended for acceptance by Special Issue on Affective Robotics. *(Corresponding author: Letícia Berto.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by The Regional Ethical Committee (Comitato Etico Regione Liguria) under Application No. IIT_wHiSPER, wHiSPER Investigating Human Shared Perception With Robots, and performed in line with the Declaration of Helsinki.

Letícia Berto is with the Institute of Computing, State University of Campinas, Campinas 13083-970, Brazil, and also with the CONTACT Unit, Italian Institute of Technology, 16163 Genoa, Italy (e-mail: leticia.berto@ic.unicamp.br).

Ana Tanevska is with the Department of Information Technology, Uppsala University, 753 10 Uppsala, Sweden (e-mail: ana.tanevska@it.uu.se).

Azamor Cirne is with the Department of Finance and Accounting, Federal University of Paraiba, João Pessoa 58051-900, Brazil (e-mail: azamorcirne@uol.com.br).

Paula Costa and Ricardo Gudwin are with the School of Electrical and Computer Engineering, State University of Campinas, Campinas 13083-970, Brazil (e-mail: paulad@unicamp.br; gudwin@unicamp.br).

Alexandre Simões is with the Department of Control and Automation Engineering, São Paulo State University, Sorocaba 01049-010, Brazil (e-mail: alexandre.simoes@unesp.br).

Francesco Rea and Alessandra Sciutti are with the CONTACT Unit, Italian Institute of Technology, 16163 Genoa, Italy (e-mail: francesco.rea@iit.it; alessandra.sciutti@iit.it).

Esther Colombini is with the Institute of Computing, State University of Campinas, Campinas 13083-970, Brazil (e-mail: esther@ic.unicamp.br).

This article has supplementary downloadable material available at https://doi.org/10.1109/TAFFC.2025.3551512, provided by the authors.

Digital Object Identifier 10.1109/TAFFC.2025.3551512

I. Introduction

A. Context

SOCIAL interaction represents a cornerstone of our everyday life, with communication, adaptation, and understanding allowing us to overcome the obstacles posed by the differences in our personalities, preferences, and perceptions [1], [2].

Our social skills enable us to recognize our peers' affective and cognitive states and, through that, also understand their needs. Through a prolonged period of this kind of interaction, we also eventually learn which behaviors are the most appropriate and well-suited for each of our interaction partners individually [3]. In this way, communication helps us gain comfort from social interactions.

However, increasing the pleasantness of our interactions with others is not the only purpose of our social abilities. Throughout our early years, communication with our caregivers develops as the first way we can make known and then eventually also satisfy our needs - when infants are hungry, they communicate with their cries, and the caregivers provide food; when they are in pain, they scream, and their caregivers try to understand the problem and help [4], [5], [6]. From these basic forms of non-verbal exchanges, children gradually develop the ability to communicate with others, laying the foundation for the intricate verbal and non-verbal interactions that characterize adult human collaboration [7].

This outlines the role of communication in two critical motivational drives in our lives - a *survival* drive, connected with the satisfaction of our base needs, and an *affective* drive, associated with the comfort we get from pleasant social interactions. A third necessary drive that guides our learning is the *exploration* drive, where guided by our curiosity, we try new feats, learn new skills, and become more self-sufficient over time. All of these drives are interlinked and are essential in our cognitive development.

These drives are not only relevant to human-human interaction but also crucial in the field of social robotics. In developing

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

autonomous robots capable of effective human interaction, it is essential to embed these drives within their cognitive architecture. Just as humans pursue their own goals and satisfy their needs while negotiating shared goals and understanding others' states and needs, robots must be designed to emulate these motivational drives. This integration is vital for creating robots that can autonomously interact with us in meaningful and adaptive ways, enhancing their functionality and our interactions with them.

This constitutes precisely the main goal of our research. In this work, grounded in human-robot interaction, we wish to investigate how different autonomous agents - humans and cognitive robots equipped with diverse motivations, value systems, and internal drives—can understand each other's needs through free-form interaction.

This study contributes to the advancement of autonomous interactive cognitive agents by exploring how drive-guided motivational systems can encapsulate social and affective interactions, thereby enhancing our understanding of adaptive robotic behavior. Our findings open new pathways for integrating emotional and social dimensions into robotic decision-making processes while preserving a functional equilibrium among competing priorities.

B. Research Background

The ability to understand the needs and affective states of our interaction partners in human-robot interaction (HRI) context is highly desirable since it paves the way for user-personalized interaction, a crucial element in many HRI scenarios - assistive or rehabilitative robotics [8], [9], [10], interactions with older adults [11], [12], [13], collaborative learning [14], [15], child-robot interaction [16], [17], and many others.

Suppose we wish for an autonomous social robot to be equipped with a motivational drive not only to explore but also to understand our needs and affective states, as well as possibly adapt its behavior continuously to them [18], [19], [20], [21], [22]. In that case, we need to explore what kind of framework for the robot may support these social functionalities [23]. We would need to endow the robot with some basic *cognitive* abilities, where sensing and perception are related to internal motivation and to an intrinsic value system to determine the proper action selection and execution while interacting with the environment and with others.

To understand the cognitive functionalities we would require in our proposed architecture, we must first grasp the concept of cognition. In our context, an autonomous cognitive agent, be it a human or a robot, is capable of *autonomously* predicting the outcome of its actions. It does so by drawing on past *memories*, current *perceptions* of the environment and other agents, and anticipation of the behavior of both the world around it and its own actions [24]. Moreover, the cognitive agent adjusts for the uncertainty of its predictions and *learns* by observing the outcome of an action and assimilating that perceptive input into its world knowledge, thus *adapting* its behavior. All of this is driven by an internal *motivational* system.

Specifically for robots in social HRI, a partial or full implementation of the essential cognitive abilities has been demonstrated in studies on autonomous social interaction [25], [26], [27], studies focused on joint action [28] or a robot assisting humans in achieving some goals [29], as well as in attention-based HRI studies on agency and dynamic behavior [30], [31], [32], where the robot's responsiveness to sensory inputs interacted with the participants' ability to infer the inherent model of operation of the robot.

There have been various cognitive architectures throughout the past few decades which have been proposed and tested on artificial agents for different cognitive tasks - SOAR [33], ACT-R [34], Sigma [35], CLARION [36], ICARUS [37] and the iCub [38] to name a few. Several exhaustive reviews of the works done in the field of cognitive frameworks can be found in [39], [40], [41].

These architectures have been developed with diverse goals, from modeling biological or psychological processes (e.g., human performance in cognitive and perceptual tasks in ACT-R or CLARION) to recreating complex cognitive processes with human-level intelligence, as in the case of SOAR or ICARUS. Our work aligns more with the latter type of architecture.

Our specific aim was to develop a cognitive architecture that could support autonomous behavior for a robotic platform, guided by various motivation drives, and evaluated in real-world interactions with humans, such as assistive or rehabilitative robotics, interactions with older adults, collaborative learning, and child-robot interaction.

In designing our study, we chose not to rely on existing comprehensive architectures, as many were too specialized to accommodate the specific drives and interaction settings we aimed to explore. A key design principle for us was scalability, enabling the future integration of new motivational drives, sensory stimuli, action ranges, and additional modules—such as those for adaptation and learning, specifically reinforcement learning (RL). This would allow for a more thorough understanding of the environment, improved behavior generalization, and enhanced interactions with various agents by better understanding also their needs.

Furthermore, integrating existing cognitive architectures with iCub would require considerable effort. In contrast, we prioritized the ability to implement a scalable system on iCub for real-world testing. Given these challenges and the limitations of existing architectures, we determined that developing a new architecture from scratch would be more effective. The proposed architecture is modular, facilitating future extensions and exploration. For example, in the context of the learning process, the implementation can utilize the Q-learning RL algorithm [42]. However, the modular design enables seamless replacement with alternative learning algorithms without requiring changes to other components. This flexibility allows researchers to explore how modifications to the architecture influence agent behavior and interaction dynamics.

C. Aim and Research Questions

The approach for designing our architecture was a developmental one, where we focused on replicating the effect of different motivation drives on the agent's interactive and adaptive capabilities.

Our architectural framework integrates multiple determinants to modulate the robot's internal emotional and behavioral states. Key factors include evaluating human peers' affective states and interaction modes, which influence the robot's internal emotional condition. Additionally, the robot's battery level is monitored to determine its survival condition, while its boredom level—affected by the availability of toys in the environment and the robot's interest in them—guides its exploration condition. These conditions collectively address the robot's fundamental needs: affective and social interaction, energy maintenance, and learning. Consequently, this architecture supports the robot in selecting appropriate behaviors, such as seeking attention and care, recharging, or exploring its surroundings. In this study, exploration was limited to the robot's field of view, which encompassed only the experimental workspace due to the robot's limited locomotion capabilities.

We hypothesize that social interaction and the ability to understand others can be based on a motivational drive influencing decision-making processes. By employing a drive-based system, we assess how the capacity to interpret social and emotional cues affects a robot's decisions. While acknowledging the critical role of social factors, we draw inspiration from Maslow's hierarchy of needs [43], [44], which classifies affection as a social need that is prioritized lower than physiological needs, such as survival. This prioritization not only aligns with our model but also underscores a key challenge in the literature: how to dynamically balance these varying needs within a robotic framework.

Our architecture was tested over its various developments on the iCub humanoid robot [45], [46], [47], [48]; and the interaction scenario we chose for evaluating the different cognitive functionalities was inspired by the interaction between a toddler and its caregiver, where the toddler is happy to explore its surroundings and learn new things. Still, when their curiosity is satisfied and they feel lonely, they then tend to seek the attention of their caregiver [49].

In our work, we want to experimentally assess how a mere change in the relative importance given to the different drives - given a fixed, simple cognitive architecture - can give rise to differences in the dynamics of interaction with human partners (acting as caregivers) and in their perception and understanding of the robot's needs. We were particularly interested in investigating how diverse robot profiles can shape the unfolding of its interaction with people and whether human partners can recognize the differences between the profiles. The robot profiles were designed with varying degrees of dependency on human partners and corresponding expectations for interaction. They were intentionally designed to explore people's understanding of the robot's needs and desires according to each profile and analyze whether they adapted their behavior to satisfy the robot's needs, assisting the robot in reaching a pleasant state.

We formulated two research questions to guide our investigation:

- *RQ1:* How do the interaction dynamics change as a function of the different robot profiles?
- *RQ2:* Can people recognize the different robot needs, and how do they evaluate the different profiles?

The rest of the paper is organized as follows: Section II presents our cognitive architecture. Next, Section III presents the experimental methods applied in our study with iCub. Sections IV, V, and VI present and discuss the study's findings, followed by a conclusion that outlines our research contributions and plans for future work.

II. The Cognitive Software Architecture

Our cognitive architecture is composed of sensory, perceptual, motivational, reasoning, behavioral, and motor modules operating simultaneously. Nevertheless, each module operates within an allocated thread time corresponding to its specific cognitive function and the time required for executing its functions. Consequently, sensory processes exhibit quicker response times than reasoning processes, exemplifying this temporal distinction. Fig. 1 illustrates the architecture.

The cognitive cycle starts by sensing the environment and the robot's internal needs via sensors. Our agent can learn its battery level, the tactile stimuli in its body, the objects in the scene, and the partner's mutual gaze and facial expressions.

Next, based on the data received from the sensing modules, our agent computes its current energy level, interest in each object within the environment, and comfort and boredom levels. These data give rise to three drives: survival (related to energy/battery), affection (associated with comfort and social interaction), and exploration (related to boredom and curiosity), which are used to motivate the robot's behavior.

Then, there is the reasoning process, represented by our decision-maker module, that is responsible for choosing which robot's needs to satisfy, giving preference for attending the most critical one. After determining the optimal behavior to perform, the agent dispatches the actions to the modules that implement the specific commands for each actuator.

The final step involves executing the chosen action. While all modules operate concurrently based on their frequency rates, the determination of the subsequent action for execution occurs immediately after the completion of the preceding behavior. This implies that actions are taken only after observing the outcomes of the prior action and assessing the most recent scenario.

Below there is a detailed description of all modules.

A. Sensory Modules

The sensory modules process stimuli from three groups of sensors: visual, tactile, and internal stimuli. Specifically:

- *Objects Detector:* Detects objects and their spatial location using color segmentation on the image captured by the camera. The only restriction is the color of the objects, which have to be black, red, green, blue, yellow, brown, or orange. The module supports using one or two cameras, and the image source is part of the data stored in each object detected. This means the robot knows which objects are seen by each camera, allowing better knowledge representation and feature extraction. We used just the left camera of iCub during our experiments.
- *Battery Detector:* Monitors the battery level, which can be obtained from either the robot's physical battery or

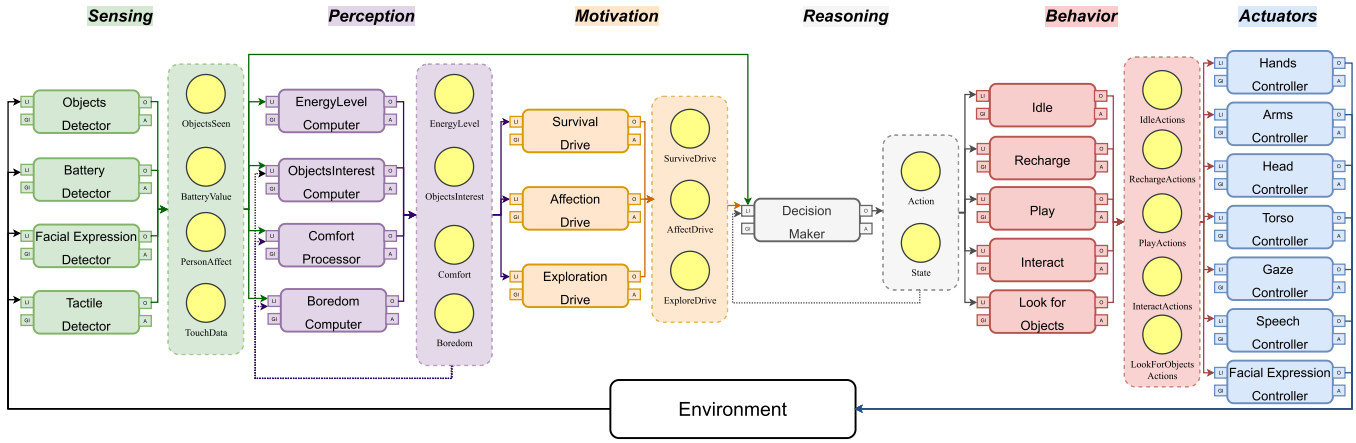


Fig. 1. The robot's cognitive cycle. Each module works at its own frequency according to its layer in cognition (sensory, perceptual, etc.) and the sensor rate. In our diagram, the data processing pathway for advanced cognitive levels follows the sequence depicted from left to right.

from a simulated one. In our case, one of the robot's tasks is learning to recharge itself, so being out of battery is an expected scenario, mainly at the beginning of the learning phase. With this in mind and given that it would be challenging to recharge the robot every time to continue the learning procedure, we used a simulated battery.

In physical robots, energy consumption varies as a function of the amount of effort required by the type of action the robot is executing. For example, walking consumes more energy than looking around. This module detects such variations using either a real battery or a simulated battery level. To simplify our experiment, all actions were assigned equal energy consumption, but these values can be easily adjusted in the simulated battery to reflect the energy demands of different activities.

Our simulated battery ranges from $[0, 100]$, with a decrease rate of 0.05 unit at each cognitive cycle and an increase of 10 units for each *Recharge* action. The energy decay rate is designed to be slow enough to prevent the need for frequent recharging during interactions yet fast enough to ensure that participants experience at least one recharging event during each interaction period. These parameters were defined through experimental trials involving the robot and volunteers.

- *Facial Expression Detector:* Extracts and analyzes the facial features of the caregiver. It detects if, in the robot's field of view, a person is looking at the robot and the person's face localization. In social interaction, people tend to display emotions via facial expressions, so this module detects the partner's displayed emotion represented by their facial action units (AUs) [50] using the open-source library OpenFace [51]. We defined positive-associated Action Units as instances of smiling and cheek raising with crinkling of the eyes. Negative-associated AUs included brow lowering, nose wrinkling, and raising the upper lip in a snarl. The classification criteria were as follows: the presence of all positive AUs was categorized as *smiling*. If only a mouth smile was present without positive upper AUs, it was considered a fake smile and not categorized as *smiling*. The presence of only brow lowering without additional negative AUs was classified as *contemplating*, while the presence of all negative AUs indicated *frowning*. The user's affective expression was labeled as *neutral* if neither positive nor negative AUs were observed in the frame [27]. Finally, if the person was not close enough to the robot to accurately detect the AUs (but was still close enough for their face to be detected), the module evaluated it as *distant*.
- *Tactile Detector:* Identifies the tactile stimuli in the robot's skin by processing data from the skin sensor patches (set of tactile sensors - aka taxels) on iCub's arms and torso. The data collected are part of the skin (left hand, left forearm, left upper arm, right hand, right forearm, right upper arm, torso) that received the touch and the average pressure of the activated taxels [52]. Additional processing was applied to the skin post-extraction due to concerns arising from prolonged interaction. The tactile sensors were susceptible to overheating, causing the inaccurate detection of minor pressure points under five taxels without actual touch, as well as not registering the touch when the pressure was very light. Consequently, the data was filtered to identify and register only those areas as touch that encompassed more than five taxels and exhibited an average pressure greater than seven, following the methods used in [27].

B. Perceptual Modules

The perceptual modules interpret the data captured by the sensory modules and give meaning to the experiences according to each module design.

- *Energy Level Computer:* Convert the battery value into a percentage representing the remaining energy available for utilization. It ranges from $[0, 100]$.
- *Objects Interest Computer:* This module plays a crucial role in continuously adjusting the interest level associated with every object the robot encounters throughout its operational lifespan. The updating process considers the object's availability and the duration elapsed since the

Algorithm 1: Comfort Typical Computation.

if $((G[t]$ **and** $F[t] \geq 0)$ **or** $T[t])$ **then**
$C[t] \leftarrow \frac{(T[t]+F[t]+G[t])+(C[t-1]\,\tau)}{(\tau+0.01)}$
else
$C[t] \leftarrow C[t-1]\beta$
end if

last interaction with each item. In each cognitive cycle, the module takes the objects observed by the robot as input and determines whether they are new or previously encountered. Upon the initial sighting of an object, it is assigned the maximum interest value. Subsequently, this interest value gradually diminishes with each interaction the agent has with the object, and the time gap between the last interaction and the current moment also influences it. Consequently, objects that have not been engaged with for an extended period are deemed more intriguing than those interacted with more recently.

- *Comfort Processor:* Within our model, iCub's sense of comfort relied on the extent of its engagement in social interactions [27], [53]. The robot experienced increased comfort when interacting with a person via visual or tactile stimulation, while the absence of any stimuli led to a decline in comfort. The comfort level range is $[0, 100]$, and in typical conditions, it is updated following Algorithm 1 Where $C[t]$ indicates the current comfort level, whereas $C[t-1]$ is the previous comfort level; $G[t]$, $F[t]$, and $T[t]$ are the input stimuli from the visual - mutual gaze and neutral/positive facial expression - and tactile sensors, respectively. β and τ represent the social variables influencing the decay and growth rates of the comfort value. In our experiments $\beta = 0.96$ and $\tau = 0.5$. The formula for the comfort and the values of the relevant parameters are defined as in [27], [53].
  When a human interacts with the robot, i.e., when iCub detects a face or registers touch on its skin, the comfort value ($C[t]$) at time t is updated using the first formula. This formula considers both interaction modalities with iCub and the previous comfort level ($C[t-1]$). Conversely, when iCub is not actively involved in an interaction, the second formula is applied to update its comfort, accounting for its decay. This formulation suggests that a multimodal interaction involving visual and tactile stimuli or a more prolonged and consistent interaction leads to an accelerated increase in comfort for iCub. The comfort dynamics were inspired by previous works on affective adaptability with robots [54], [55].
  However, in two exceptional cases, comfort is computed differently. The first case is when iCub is recharging (action done by sleeping), so there is no visual input or tactile conscious processing, so the comfort processor uses the second formula but with the modification of $\beta = 0.98$ to ensure the comfort will not decay too quickly. The second case is when the robot stays at the minimum comfort level for approximately 75 seconds or longer, and the person does not interact with the robot, even if the robot attempts to get attention and care. So, the iCub must find a solution to cope with the shortage of attention, and in this case, it looks at a picture (iCub and some people playing together) available in the scenario and takes it as a good memory, providing a small increase in comfort. In such a situation, the formula used is $C[t] = (0.5 + (C[t-1]\tau))/(\tau + 0.01)$ and it is used for the next 20 module's frame. This strategy was inspired by children's behavior, which is usually to seek comfort by playing with their favorite (or the available) toy in absence of the caregiver's response [56], [57].
- *Boredom Computer:* We modeled boredom as an internal value, an inner state of mind. Typically, the boredom increases 0.1 per timestep, but if the robot is recharging, the increase rate is slower (0.02), trying to simulate what happens with a child who wants to play but has to rest before. The solution to reducing boredom is playing with the toys available in the scenario, decreasing the value according to the interest value (maximum 10 in our experiments) of the object the robot plays with during the interaction period. The boredom level ranges from $[0, 100]$.

The parameter values outlined above were determined through extensive testing, including pilot trials with volunteers. The pilot tests were conducted in two sessions. In the first session, participants were asked to perform specific behaviors with the robot to verify that the timing and robot reactions aligned with the design expectations. This phase allowed us to fine-tune the parameters to ensure that the dynamics of state changes were appropriate. For instance, if the robot requests interaction and detects a complete lack of attention, it waits 20 seconds before explicitly asking for interaction again. In the second session, conducted only after all the pilot testers had completed the first phase, participants engaged in free-form interaction to validate whether the adjustments made led to more realistic and improved interaction dynamics.

C. Motivational Modules

Our cognitive robot has three needs, which we refer to as survival (interoceptive), affection (social), and exploration (internal feeling). Based on the concept of homeostasis [58], an imbalance in these needs gives rise to drives that motivate behavior.

For each need, we compute the related drive by the difference between the current need value and the homeostasis level associated with this need. The homeostatic level is determined by the combination of the homeostatic value (H) and a range value (R), which establishes the extent of the optimal zone ($[H - R, H + R]$) for the agent. It represents the agent's equilibrium state, and these values can be changed if necessary to test other hypotheses and explore other behaviors. Furthermore, it can be different for each need. In our experiments, each need has its own homeostasis and range values varying according to the robot profile.

Homeostasis does not equate to maximum need fulfillment; for instance, humans do not require continuous social interaction. Needs can exceed (saturation, e.g., social fatigue) or fall below (critical levels, e.g., loneliness) homeostasis, triggering

neutral, positive, or negative drives, respectively. To mirror these dynamics, our artificial agent incorporates similar mechanisms. Consider energy needs as an example: children may feel tired or hungry, but when enjoying time with friends, they often resist stopping to rest or eat. In these moments, the joy of playing outweighs their immediate need for rest or nourishment, making continued play feel more rewarding than going to sleep.

Therefore, the drives active in our framework motivating iCub are:

- *Survival Drive:* It is related to energy needs and consequently surviving. It can motivate behaviors like recharging when the drive is negative or just consuming energy when it is positive.
- *Affection Drive:* It is related to affect and social interaction needs based on comfort level. It gives rise to behaviors like asking for attention and care and trying to interact with others when the drive is negative or disengaging social interaction and trying to be alone when the drive is positive. In this work, we modeled the affective drive in a simplified manner, focusing on participants' physical interactions and expressed emotions to explore how such social stimuli shape interactions. This simplified approach, proven effective in the context of HRI [53], introduces affective elements into human-robot interaction, serving as a foundation for understanding the dynamics of social engagement. Despite its simplicity, we are integrating this framework with broader considerations such as autonomy and exploration, offering a novel perspective on the intersection of social and cognitive robotics.
- *Exploration Drive:* It is related to the desire to try new things caused by boredom. This drive can energize behaviors like playing with toys in the environment when the drive is negative or taking a pause of playing and starting new activities when the drive is positive (due to loss of interest, prolonged repetition of the activity, and sense of being unable to explore new approaches to the task).

Within this framework, our agent's motivations are shaped by a range of factors, including intrinsic characteristics, the surrounding environment, and interactions with partners, each affecting the fulfillment of these motivations to varying degrees—from complete dependence on the agent itself to complete dependence on external partners. Specifically, the affection drive relies entirely on human interaction, while the exploration drive is influenced by the partner's engagement with toys, the availability of toys in the environment, and the agent's own boredom levels. In contrast, the survival drive depends solely on the agent's internal mechanisms for fulfillment.

D. Reasoning Modules

The decision-making module handles reasoning in our cognitive architecture, using logical interpretation based on the robot's context to make choices. This module considers only the drive values, so it receives all their values as input and chooses the most critical one (the most negative value) to satisfy. Additionally, a priority schedule is established in case of a tie, ranked from highest to lowest as follows: Survival, Affection, and Exploration. In this manner, the module dynamically establishes priorities and employs a dynamic action selection process, selecting the optimal action based on the robot's motivations, context, and available knowledge at that particular moment.

E. Behavioral Modules

Given the decision, the behavioral modules are responsible for choosing the series of actions required to satisfy the unbalanced drive. In our scenario, iCub is a baby robot with limited movements and only speaks keywords and babbles. The robot speaks simple keywords for all the behaviors to improve its partner's understanding of what it is doing.

- *Idle:* The action executed is looking around and whistling occasionally. This behavior is the best to execute when all drives are above homeostasis (positive), given that it contributes to consuming energy, reducing the comfort level (as the robot is not looking at the person), and possibly encountering new things/people in the environment that can be interesting (like a baby that from time to time looks around observing the surroundings).
- *Recharge:* Say "sleep" and assume the sleeping position (eyes closed, arms relaxed, and snoring). As we are using a simulated battery, this module sends a signal to the battery sensor to recharge (if it was using the genuine battery, the action should be moving to the recharge station) and executes this action until it reaches the energy homeostasis. When this action is finished, the robot yawns and opens its eyes.
- *Look For Objects:* Look at the table to check if there are toys.
- *Play:* Say "play", then look and point to the object chosen while babbling. In our scenario, the robot chooses the objects to play with according to the highest interest value. In case of a tie between objects, the robot chooses the one seen first.
- *Interact:* Asks verbally for caring while raising the arms in the person's direction (as a baby does). If the person does not reply, iCub starts to cry and tries to get its partner's attention. If, after a long time, the behavior is still *Interact* (because the comfort level is critical and there is no other drive more urgent), iCub looks at a picture with him and his "friends", sighs, and look again at the person. While executing this behavior, if the partner touches the robot, it looks at the touched part and sometimes laughs.

F. Motor Modules

The motor modules are responsible for implementing the specific commands required for each actuator. They are:

- *Hands Controller:* Right and left hands movement.
- *Arms Controller:* Right and left arms movement.
- *Head Controller:* Control the head and neck.
- *Torso Controller:* Move the torso back and forth.
- *Gaze Controller:* Move the eyes, head, and neck.
- *Speech Controller:* Audio system and text-to-speech.
- *Facial Expression Controller:* Facial LEDs (eyebrows and mouth).

It is important to highlight that iCub can perform smooth and natural motions, but in our study, it was presented as a toddler, so it has simple and limited movements. Moreover, we opted to minimize the linguistic abilities of the robot to mimic the competencies of a preverbal child, to focus on an interaction mediated by actions and non-verbal signals.

The complete infrastructure, including all modules, sensors, connections, and software required to integrate our cognitive architecture with the iCub robot and replicate the experiment, is provided in the supplemental material.

G. Demonstrating Architectural Dynamics

The cognitive architecture allows the robot to operate autonomously in the environment, considering the stimuli received and its internal state. It allows iCub to interact appropriately with anyone without previous knowledge or contact with them and without the experimenter's intervention.

We propose that adjusting a single parameter within the cognitive architecture is sufficient to initialize different agents. This idea is supported by embodied cognition theories, which suggest that cognition emerges from the interaction of the body with the environment. Subtle changes in internal states—such as motivations, drives, or emotional states—can significantly influence perception and behavior without the need for a complete redesign [59], [60]. Merleau-Ponty's concept of embodied interaction further supports this, highlighting the role of internal states in guiding behavior within specific environmental contexts [61]. Therefore, by fine-tuning parameters related to the functions governing attention orientation, social preferences, and need for energy, robots can exhibit diverse behaviors, demonstrating the flexibility and scalability of the architecture. With this perspective, analyzing the consequences and unfolding interactions caused even by minor modifications in particular modules becomes feasible. This approach opens up a spectrum of possibilities, including exploring the effects of various combinations of modules and the connections established and learned among cognitive functions.

Hence, we used the same cognitive architecture and motivations but varied the agent preferences by changing its homeostasis levels, resulting in two profiles. In the first profile, the robot prefers social interaction via touch and visual contact, driven by the need for increasing comfort. In contrast, the second profile prefers playing with the toys on the table and environmental exploration with a higher drive to reduce boredom. Despite the presence of preferences exhibited by the robots, the motivations are contingent upon the dynamics of interaction, which are influenced by the nature of partner engagement and iCub's internal states.

In the subsequent sections, we will refer to the first profile as "social", while the second is "playful". We initialize the robot with positive drives, i.e., above homeostasis, and a negative imbalance in the drive corresponding to the robot profile. So, for the social one, comfort starts at the minimum level, being the most critical drive, making the robot start the interaction by asking for care and attention. For the playful one, the robot starts with a maximum boredom level (critical), so the robot starts wanting to play. The subsequent behaviors are entirely contingent upon the interaction between the iCub and its partner.

Fig. 2 shows the architecture dynamics and the trends of iCub internal values and decisions in three different interaction sessions. Each need is defined within a homeostasis range tailored to the robot's profile, impacting the trigger period for respective drives. The further away from homeostasis in the negative direction, the more stressful the situation for the robot.

Fig. 2(a) and (d) portray the behavioral dynamics observed when a participant initially interacts with the Social robot profile. This participant exhibits the expected behavior by responding appropriately to the robot's cues and engaging in tactile and visual interactions (Fig. 2(g)) as prompted by the robot's demands. Notably, the participant adjusts their interactions based on the robot's activities, providing stimuli during interactive phases and refraining from such actions during the robot's play sessions. Although occasionally leaving the robot to its explorations while idle, this participant demonstrates a clear understanding of its needs and adapts their behavior accordingly to facilitate a harmonious interaction, thereby aiding the robot in achieving a pleasant state, as can be seen in Fig. 2(a) where the needs, and in particular the comfort and boredom, are conducted in the homeostasis range (or above) for most of the time.

Fig. 2(b) and (e) depict interactions between iCub and a highly interactive individual during their session with the Playful profile. Throughout this session, the robot seldom experiences negative Affection drive values and maintains consistently high comfort levels. Analysis of Fig. 2(h) reveals the participant's frequent physical contact with the robot, even during non-interactive phases, indicating an abundance of stimuli provided. The participant exhibits predominantly neutral to positive facial expressions throughout the interaction and also sustains visual engagement, seeking mutual gaze independent of the robot state, displaying unusual effort to maintain contact with iCub. However, the robot remains idle for extended periods due to excess stimuli and having all the other drives satisfied.

Conversely, Fig. 2(c) and (f) illustrate iCub's internal states during an interaction with a participant who inadequately supplies sensory information in their session with the Playful profile. Fig. 2(i) highlights the participant's limited physical interaction and lack of visual engagement with the robot throughout the majority of the interaction, only offering minimal stimuli when prompted. Consequently, the robot predominantly remains in the Play state, as anticipated by its profile, and the Interact state due to insufficient attention from the participant. Towards the conclusion of the interaction, iCub experiences prolonged discomfort, caused by minimal comfort values, prompting corrective actions done by itself, such as looking at the picture that retrieves a lovely memory to address the critical drive deficiency temporarily. Concerning the Survival drive, it remained stable for most of the interaction, only starting to turn negative around time 480, indicating mild fatigue. However, as two other drives were more urgent—one being in a critical state—the robot prioritized addressing these needs, similar to children who, despite feeling tired, don't want to stop spending time with their friends. If the survival drive had reached a level of urgency equivalent to the

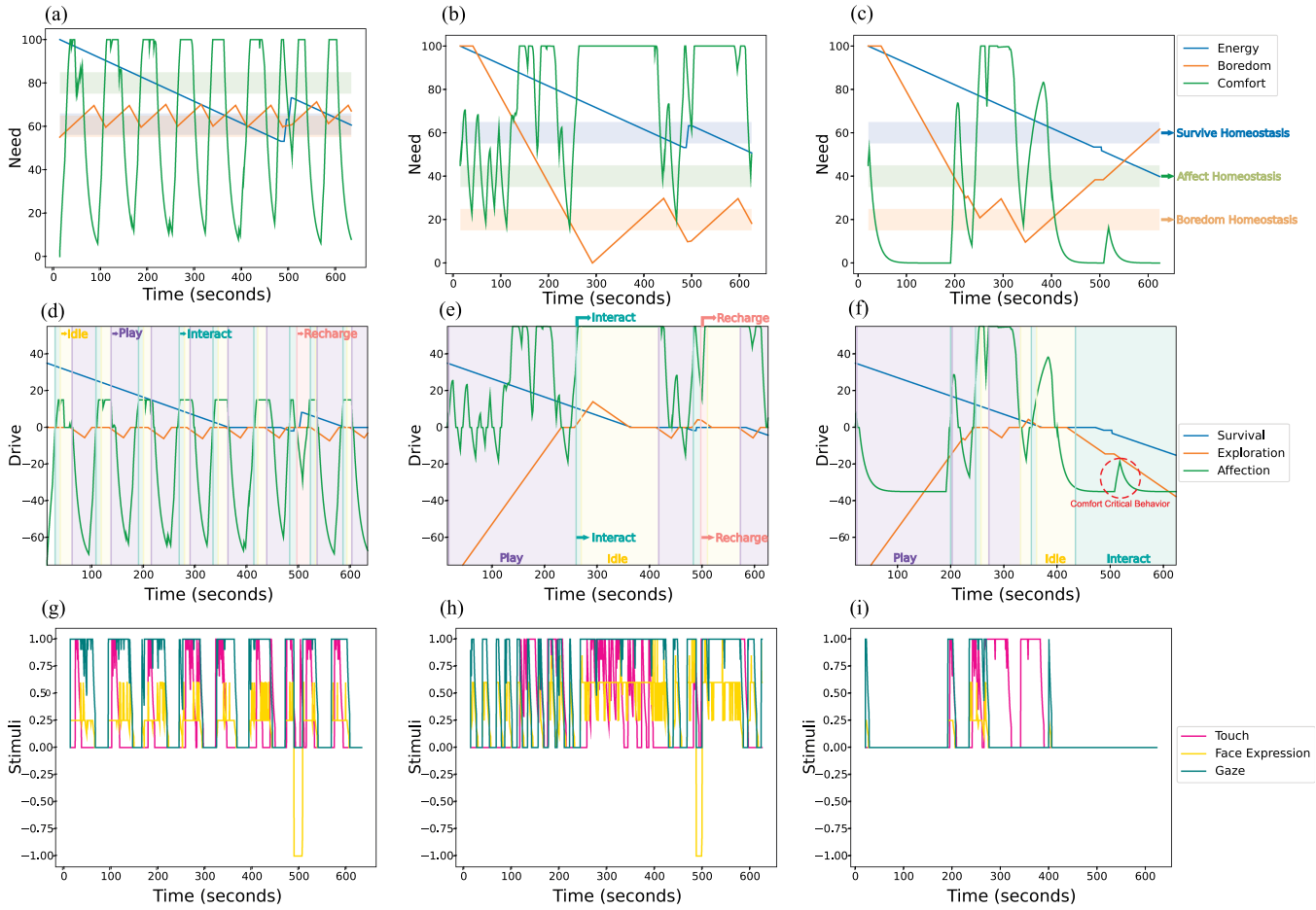


Fig. 2. Dynamics of iCub's internal states and external stimuli during interaction sessions. The upper graphs illustrate variations in the robot's energy, boredom, and comfort levels. In contrast, the middle graphs depict the dynamics of Survival, Exploration, and Affection drives, considering the associated need values and their homeostasis ranges defining the robot behavior decisions. The lower graphs detail the occurrence of sensory stimuli. (a), (d), (g) depict a sample of a participant interacting with a Social robot in the first session, demonstrating expected behavior and providing appropriate stimuli. (b), (e), (h) illustrate a sample of a participant interacting with a Playful robot in the first session, showing an overabundance of input stimuli, leading to periods of robot idle state. (c), (f), (i) present a sample of a participant interacting with a Playful robot in the second session, where insufficient interaction resulted in prolonged discomfort for the robot, prompting urgent action to address the drive imbalance.

other needs, the robot would have opted to recharge, adhering to the predefined priority hierarchy in case of equivalent urgency.

III. Materials and Methods

A. Experimental Design

Our experimental setup consisted of a robot positioned in a fixed place in the laboratory, a chair for the participant, and a table full of toys between the robot and the participant, as illustrated in Fig. 3. The experiment was explained in Italian and English, and the iCub was described using the pronoun "him"/"lui" (in Italian). The context was presented as a child-caregiver interaction to propose an interaction setting simple enough to be autonomously and effectively handled by the robot but still enabling rich interaction with the human partner, where the effect of the different robot profiles could emerge in the robot behavior.

The story told to all the participants was that iCub was a young child/baby, unable to talk but trying to communicate by



Fig. 3. A participant and the robot iCub during the interaction.

emitting sounds. Also, the robot had its own needs and needed the partner's help to satisfy some of them. The robot could feel lonely or want to be sure that someone is with him so he can ask for care. Moreover, it was explained that iCub likes to discover more about what the toys could do and how to play with them, but he did not yet know how to manipulate them, so he could

point to the object of interest and expect the partner to interact with the toy showing to him the possibilities. Also, as a child, iCub could feel tired; in these moments, he needed to rest and recover his energy. Sometimes, he was just interested in looking around the environment to see if something was enjoyable or new for him like we do when idle.

We explained to the participant that the task involved taking care of iCub when we were not there and acting as caregivers. They could do what they wanted in a free-form interaction, attending (or not) to the robot's needs and in the form they felt more comfortable with. We also told them that the robot could feel touches on the arms and torso, see the participant, and detect if they were happy, and that iCub could see the objects.

We opted to present the robot as a baby to simplify the understanding by the participant of the limited range of action possibilities we were focusing on in the current setting: our robot did not need to walk, skillfully manipulate objects, or do natural language processing. Besides that, having a baby robot demands care from someone, and the participant could understand the task without previous experience. Also, as we work on incremental complexity in cognitive capabilities and motor development, having a baby is the ideal agent.

We built a child-oriented and colorful scenario with different kinds of toys to effectively evoke the atmosphere of a typical child's surroundings. The toys were easy to manipulate, and the participants could explore diverse aspects, like sounds, texture, and movements, besides creating new toys with Lego.

B. Participants

In total, 36 participants (17 female, 19 male, avg. age 26.1, SD 4.99) from five countries (Italy, Germany, France, Lebanon, and Egypt) interacted with both robot profiles in a balanced order without being informed of the difference. This resulted in two groups: PS, referring to participants who interacted first with the Playful robot and then the Social, and SP, for the participants in the inverse order. All participants gave written informed consent before participating and received a fixed refund of €12. Regione Liguria's regional ethics committee approved the experimental protocol.

To account for the potential effect of the order in which participants encountered the two profiles, we counterbalanced the order. Consequently, the participants were allocated into two groups, each comprising 18 individuals. In one group (SP), participants engaged initially with the social profile followed by interaction with the playful profile, while in the other group (PS), the sequence was reversed.

C. Procedure

Upon arrival at the institute, participants were directed to the laboratory room hosting the iCub robot, which features a dedicated area, delimited by a sliding door, creating distinct control and experimental areas (Fig. 4). In the control area, the experimenter remained throughout the study without direct visibility of the participant and iCub. This deliberate arrangement aimed to foster participant comfort with iCub, mitigating concerns about judgment that might hinder natural interaction. Additionally, it

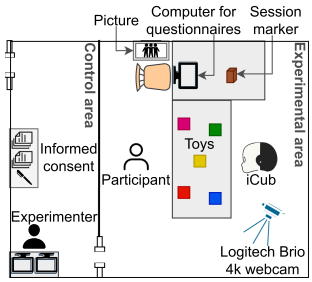


Fig. 4. The experimental setup.

enhanced the narrative's realism, portraying individuals as caregivers for the robot in the experimenter's absence. In the control area, participants executed the informed consent process, stored their belongings to minimize distractions during the experiment, and were introduced to the study procedures.

Subsequently, the experimenter guided the participants to the experimental area, where the robot was sleeping with closed eyes and simulated snoring. Here, the experimenter provided an overview of the robot's tactile sensing capabilities, highlighted sensor positions for optimal interaction, and encouraged participants to touch the robot. Explanations covered the robot's visual perception of individuals and environmental objects, its restricted range of movements, and its ability to listen but without good verbal communication despite the ongoing learning and development efforts. Following the initial interaction with the sleeping robot, participants relocated to a secondary table with a laptop to complete an initial questionnaire (PRE). During this interval, the experimenter returned to the control area and awaited the participant's readiness.

Upon questionnaire completion, participants assumed a seated position in front of the robot and a table adorned with toys. The formal experiment was initiated when participants signaled their readiness. iCub gradually awakened, opened its eyes, yawned, and directed attention toward the participant, introducing itself as "Berry" while waving its hand. Participants were unaware of two distinct robot profiles and were informed they would engage in two interactions (separated by a questionnaire) with the robot. The robot's appearance, including accessories, name, voice, and other features, remained identical to preclude potential cues from revealing any differences. The primary objective was to examine participants' ability to discern disparities emerging from the interactions. Each interaction was systematically recorded from iCub's perspective (left eye camera) and an external camera.

An interaction session in either profile setting lasted 10 minutes. In addition to interacting with the robot, participants completed questionnaires (additional information is available in the subsequent subsection). This supplementary activity increased the total time commitment for participants to approximately 40–45 minutes.

D. Measures

During the experiment, we collected data from 3 distinct sources: the questionnaires filled by the participants, the data

collected by the robot during the interaction, and video recordings captured by an external camera.

1) Questionnaires: All questionnaires were presented in Italian, using Survey Monkey,[1] and administered to participants at four distinct stages of the experiment. In all the questionnaires, participants were asked to insert a personal anonymous code, self-created according to our indications, to connect the four questionnaire answers. We asked participants to answer all items on a 7-point Likert Scale.

The initial set of questionnaires (PRE) was conducted after participants entered the experimental area, observed the robot dormant, and received enough information about the robot before engaging in a real interaction. Within this set, we gathered data on self-perception and perception of the robot.

In *perception of the self*, we collected demographics data, caregiving experience, previous experience with iCub, and a brief measure of the Big Five personality traits (extraversion, agreeableness, conscientiousness, emotional stability, and openness to experience) [62].

Regarding *perception of the robot*, participants provided impressions on the robot's anthropomorphism [63] (i.e., the extent to which they perceived the robot as resembling a human being), likeability (i.e., how much they liked iCub) - the scale was partially adapted from [64], following what was done by [65] in an HRI context, warmth and competence [66] (i.e., the extent to which they perceived the robot as both warm and competent), and agency and experience [67] (i.e., the degree to which they believed the robot could act and feel). Furthermore, we employed the Inclusion of Other in the Self (IOS) questionnaire [68] to understand participants' sense of closeness to the robot.

The subsequent set of questionnaires was administered following the initial interactive session with the robot (PLAYFUL for the PS group, SOCIAL for the SP group), aiming to gather data on the evaluation of the robot using the same questions as in the PRE questionnaire. Additionally, participants were presented with a qualitative open-ended question asking them to describe the robot's personality using five words.

The third set of questionnaires took place after the second interaction with the robot (SOCIAL for the PS group, PLAYFUL for the SP group). It mirrored the questions in the second set, except that participants were required to evaluate the robot based on the second robot profile.

The final set of questionnaires (END) was administered at the conclusion of the experiment. Within this phase, data were collected to determine participants' awareness of differences between the interactions, identify the robot's most demanding aspects in both the first and second sessions, and ascertain participants' preference for one session. This questionnaire was designed to assess and evaluate the overall effectiveness of the interactive sessions.

The complete questionnaires are provided in the language they were administered (i.e., Italian) and in English within the supplemental material.

2) Internal Data From iCub: Data acquisition from iCub involved recording data captured by various sensors, including visual, tactile, and joint sensors, as well as internally computed values within the architecture. These internal values encompassed the percentage of battery level, detected objects and the robot's respective levels of interest, the presence of individuals, mutual gaze and expressed emotions, and comfort and boredom metrics. Additionally, the recorded data included information on drives, actions performed, and the temporal intervals between each decision and behavior execution. Each frame received by the robot was meticulously annotated with a timestamp, ensuring precise temporal alignment for subsequent analysis.

IV. Results

Considering the different aspects of our data collection, we present the results in two macro-categories: (I) subjective, regarding data from questionnaires, and (II) behavioral evaluation, regarding data from the interaction with iCub. All the statistical analyses were conducted using Jamovi[2] and SigmaPlot.[3] Before conducting all tests for mean comparisons, preliminary assessments were performed to verify the assumptions of normality and homogeneity of variances. After meeting these assumptions, parametric tests and post-hoc analyses were conducted when statistically significant differences were observed. In cases where at least one assumption was violated, appropriate non-parametric tests and corresponding post-hoc analyses were utilized contingent upon statistically significant differences.

A. Subjective Evaluation

The subjective evaluation explores how each robot is evaluated according to the participant's perception and experience of interacting with it and the participant's ability to differentiate between the two robot profiles. It investigates whether their impressions of the robot's anthropomorphic characteristics, likeability, warmth, competence, agency, experience, and IOS ratings change relative to the time spent with the robot or specific robot profiles.

Initially, items formulated negatively to prevent participants from agreeing with them were reversed appropriately. Subsequently, for each scale, we computed the descriptive statistics and internal consistency using Chronbach's α [69]. All the scales with $\alpha > 0.6$ were considered consistent [70]. The complete descriptive statistics, including mean and standard deviation, and Cronbach's α results, are available in the supplemental material.

In our study investigating how the iCub profile influences participants' perceptions of the robot and their sense of closeness, we examined participants' average ratings across three phases: before interaction (PRE), and after interacting with Playful and Social robots.

We first sought to explore the potential impact of interaction order on participants' evaluations of the robot. Consequently, we conducted a comparative analysis of results across different interaction sequences. The Shapiro-Wilk test revealed normal distributions for all variables except for the Experience and IOS scales across all three phases. Accordingly, T-test (parametric)

[1] https://it.surveymonkey.com

[2] https://www.jamovi.org

[3] https://grafiti.com/sigmaplot-detail/

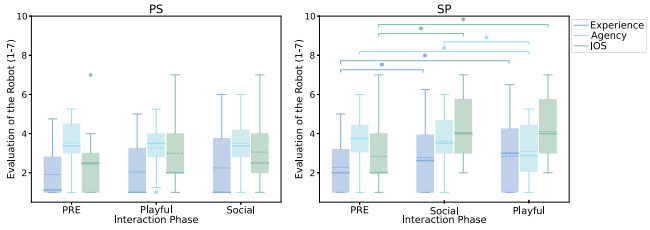


Fig. 5. Experience, Agency, and IOS ratings average across order and phases. Asterisks indicate significant differences. The order of conditions is depicted according to the temporal order in which they have been faced by each group.

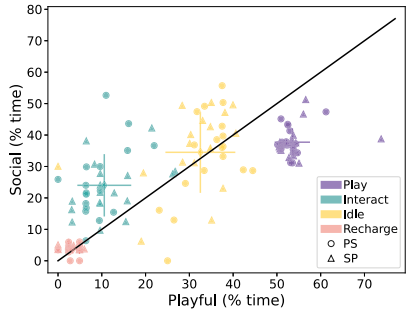


Fig. 6. Relationship between participants' time allocation in the main interaction states across both robot profiles. Crosses represent the averages for each state.

and Mann-Whitney (non-parametric) tests were utilized. Our findings indicate no significant disparities between groups before the interaction phase or for the social profile. However, a statistically significant difference $[U = 97.00; p = 0.037]$ was observed in the IOS scale specifically for the playful robot profile, with the SP group having higher IOS than the PS.

Then, we analyzed how the evaluation changes during each phase for each group. We first assessed the data's normality using a Shapiro-Wilk test, revealing non-normal distributions for the Warmth, Likeability, Experience (for the PS group only), and IOS scales. We then conducted statistical parametric analysis tests for normally distributed scales to assess variations between rating phases using ANOVA. In contrast, non-normally distributed scales were analyzed using the non-parametric Friedman test.

Results showed no statistically significant differences (all $p > 0.05$) between phases for the PS group. However, for the SP group, significant differences were found in the Experience $[F(2; 34) = 4.535, p = 0.018]$, Agency $[F(2; 4) = 6.059, p = 0.006]$, and IOS scales ($p < 0.001$). Posthoc analysis using the Holm-Sidak method revealed that the Experience attribution increased significantly from PRE to Social ($p = 0.041$) and from PRE to Playful ($p = 0.027$), while Agency decreased significantly from PRE to Playful ($p = 0.007$) and from Social to Playful ($p = 0.025$). Regarding IOS, Friedman $[\chi^2(2) = 14.978; p < 0.001]$ and a Tukey posthoc test indicated a significant increase from PRE to Social ($p < 0.05$) and from PRE to Playful ($p < 0.05$). Fig. 5 succinctly summarizes the statistically significant differences in these scales for both groups.

In the subsequent END questionnaire, participants were queried regarding their perception of disparities across sessions, with 35 respondents affirming the presence of distinctions. Using a 7-point Likert scale to evaluate the perceived primary demand of each robot, the results were as follows (mean [standard deviation]): For the playful profile - Play: *6.25* (0.8), Sleep: 3.42 (1.4), Caregiving: 4.11 (1.7); and for the social profile - Play: 4.47 (1.3), Sleep: 3.28 (1.7), Caregiving: *6.14* (1.1). These findings highlight participants' ability to recognize the distinct preferences associated with each profile.

In conclusion, regarding participant preferences for the robot profiles, the playful profile garnered preference from 58.33%, the social profile from 22.22%, both sessions were equally enjoyed by 11.11%, and 8.33% found neither session enjoyable. Although we did not specifically investigate why participants preferred the Playful iCub over the Social one, analysis of the word cloud and post-experiment feedback revealed that the session with the Playful profile was perceived as more interactive and enjoyable, allowing both partners to engage in fun activities together. In contrast, the Social profile sessions left participants feeling more like caregivers, responsible for addressing the robot's discomfort. We performed a statistical analysis to examine the correlation between personality traits and participants' robot preferences; however, no significant differences were found.

B. Behavioral Evaluation

In this section, we analyze the data gathered by the robot during interactions. As indicated in Section IV-A, most participants reported their ability to discern differences between the two robot profiles and accurately assess each robot's primary demands. Consequently, we investigate participants' interaction patterns with the robots and ascertain whether they adapt their interactions.

1) Time Distribution Across States: The initial analysis aimed to examine the distribution of time spent by each participant in the main states for each robot. Fig. 6 illustrates that, as anticipated, the proportion of time allocated to the *Recharge* state is notably low for both robot profiles. Conversely, the duration of time spent in the *Idle* state exhibits considerable variability. Upon closer examination of the data, it was observed that lower values corresponded to instances where participants provided insufficient stimuli to the robot, resulting in prolonged periods in either the *Interact* or *Play* state. Conversely, higher values indicated scenarios where participants provided ample stimuli to the robot, satisfying its drives for an extended duration. The distribution of the *Play* and *Interact* states is concentrated below and above the identity line, respectively. This pattern indicates that participants tended to engage the robot for longer durations in a state that aligned with its inherent profile characteristics.

Following this analysis, we examined the average behavior of the PS and SP groups, as depicted in Fig. 7. The results for both experimental groups were similar. As anticipated, the Playful robot spent more time playing than in other states. On the other hand, the Social robot spent more time playing or idle rather than requesting interaction. This can be attributed to participants

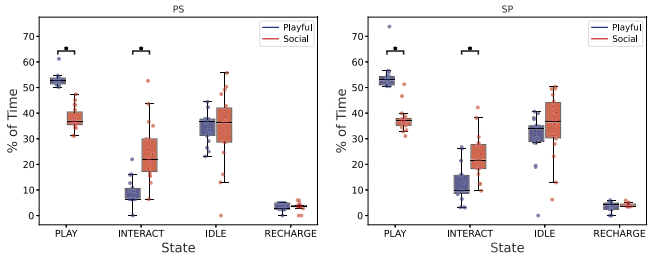


Fig. 7. Average percentage of interaction time spent in the primary states, depicted across sessions.

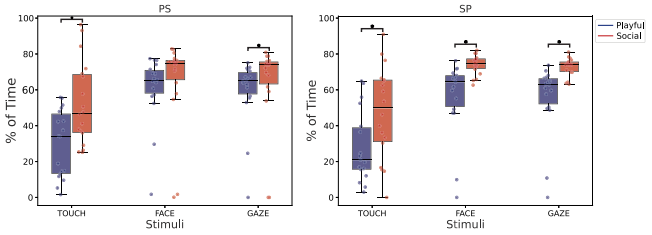


Fig. 8. Average percentage of time the robot perceived each input stimuli, depicted across sessions.

touching the Social robot more frequently, perceiving it as more needy, thus reducing the need for it to request attention and increasing its time spent in other states. Although the Social robot spent considerable time in Play, it was less than the Playful robot.

Regarding the time spent in the "Idle" state, we observed that the Social robot tended to spend more time there compared to the Playful robot. This observation is consistent with the increased physical interaction (touch) and visual engagement (gazing at the person) noted in the social condition. These interactions enhanced the robot's comfort level more rapidly and, considering the dynamics of other motivational drives, the robot achieved a state of pleasure and satisfaction more frequently. Additionally, the social profile exhibited more interactive and idle states variability than the playful profile.

Regarding the "Recharge" state, both robot profiles spent minimal time in it. This was designed to prevent the robot from being inactive for extended periods, which could potentially cause participants to become disengaged or uncertain about how to proceed with the interaction.

Despite the robots having distinct preferences, they were designed to enter each state at least once, ensuring that all participants experienced all the robot's behaviors. Considering this design aspect and the observed results, we compared the time spent in each state for each robot profile. A Shapiro-Wilk test for normality indicated that the data were non-normally distributed ($p < 0.05$), except for the Interact state in the SP group ($p = 0.115$). Consequently, we employed the Mann-Whitney test with Yates continuity correction for the non-parametric data and Welch's t-test for the parametric data. The results indicated a statistically significant difference between the Playful and Social profiles in the time spent in the Play ($p < 0.001$) and Interact ($p < 0.001$) states for both groups. We found no significant difference in the time spent in each state for the same robot profile between the two groups, showing the order in which participants encountered the two profiles did not affect how their time was distributed among the different states.

2) Stimuli Perception Across States: The state the robot executes is defined by its internal drives that partially depend on the participant's stimuli perceived by the robot. So, in Fig. 8, there is the average time the robot perceived each stimulus. An important observation is that a lack of input stimuli can trigger critical comfort behavior when the robot is in the *Interact* state. Only two participants caused the robot to execute this critical comfort behavior. However, following the robot's action, they altered their interaction and apologized to the robot.

In the PS group, the Playful robot spent less time perceiving touch stimuli than the Social robot, which exhibited a broader distribution and higher median values. The box plot for the Playful robot indicates the presence of outliers, suggesting a few sessions with notably low touch interaction times. Both robot profiles spent a similar percentage of time perceiving face stimuli, with overlapping distributions and close median values, indicating low variability within each profile. The time spent perceiving gaze stimuli was consistent for both profiles, though the Social robot had slightly higher median values.

In the SP group, the Social robot spent more time perceiving touch stimuli than the Playful robot, mirroring the trend observed in the PS group. The Playful robot's touch perception exhibited higher variability with a broader distribution. The distributions for face stimuli perception were relatively similar between the two profiles, with the Social robot showing a slightly higher median. The time spent perceiving gaze stimuli was very similar for both profiles.

Regarding statistical analysis, a Shapiro-Wilk test indicated that the data were distributed non-normally ($p < 0.050$), except for the touch stimuli in the SP group. Consequently, we used the Mann-Whitney test for non-parametric data and t-test for parametric data. In the PS group, we found a significant difference between the Playful and Social profiles for the touch ($p = 0.013$) and gaze ($p = 0.028$) stimuli. In the SP group, significant differences between the two profiles were found for all stimuli: touch ($p < 0.05$), face ($p < 0.05$), and gaze ($p < 0.001$).

Overall, the Social robot generally perceives more touch stimuli than the Playful robot across both groups, indicating higher engagement through physical interaction. The differences between the two profiles regarding face and gaze perception are minimal, suggesting that participants perceived these stimuli similarly important for both profiles. The Playful profile exhibits more variability in touch perception, particularly in the SP group, whereas the Social profile maintains more consistent interaction times across all stimuli.

These findings suggest that the Social robot elicits more consistent and frequent physical interactions (touch), whereas the Playful robot's interactions are more variable. The similar patterns observed in face and gaze perception indicate these stimuli are less influenced by the robot's profile.

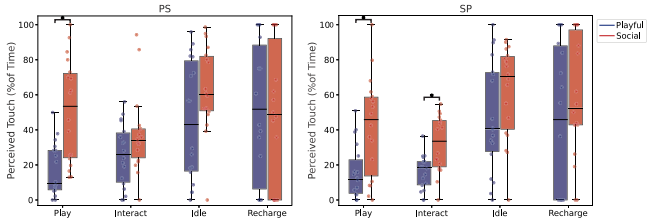


Fig. 9. Average percentage of time the robot perceived the touch stimuli in each main state, depicted across sessions.

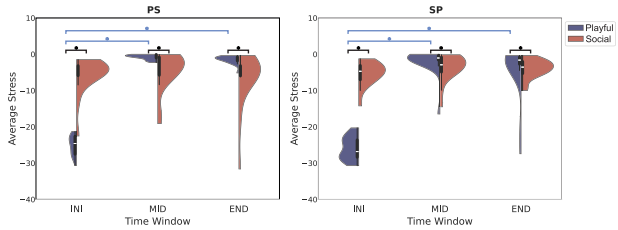


Fig. 10. Robot's stress levels measured at three points during the interaction: beginning (INI), middle (MID), and end (END).

3) Differences in Participant Behaviour: Furthermore, our investigation aimed to discern participants' behavior, particularly their responses to the robot's various states, to ascertain whether they comprehended the robot's requirements at each state, even without explicit instructions. We focused solely on touch stimuli (depicted in Fig. 9) because the occurrence of face/gaze stimuli also depended on where the robot directed its cameras. This differed between profiles, with the playful robots looking more at toys and away from the partner's face.

We conducted our analysis in two distinct categories: (I) intra-group, where we compared interactions within each group according to their order, and (ii) inter-group, focusing on comparing interactions with each robot between the groups. In the intra-group analysis, participants in the PS group demonstrated an increased frequency of touch during the Play, Interact, and Idle states in their second interaction (with the social robot), while maintaining similar touch frequencies during recharge. Conversely, participants in the SP group exhibited reduced touch frequencies during the same states in their second interaction (with the playful robot), with touch frequencies during recharge showing higher variability but generally decreasing. These findings suggest a general understanding of the robots' profiles, with participants adjusting their interactions accordingly, even without explicit instructions. This behavior underscores a caring attitude towards the robots, engaging in behaviors conducive to long-term interactions, such as showing greater care towards the social robot even when not explicitly prompted.

Regarding the statistical analysis comparing the percentage of time spent touching in each state between the two robot profiles, we found the following significant differences: (I) For the PS group, there was a significant difference only in the Play state ($p < 0.001$), and (II) For the SP group, significant differences were observed between the two profiles in both the Play state ($p = 0.002$) and the Interact state ($p < 0.05$).

In the intergroup analysis, concerning the playful robot, the PS group exhibited a higher frequency of touch during the Interact state and displayed greater variability during Idle. This difference could stem from the SP group having learned from their initial interaction with a robot that frequently seeks attention but stops when it wants to play, whereas the PS group might not have been aware of this pattern yet. Conversely, for the social robot, the PS group engaged in more touching during the Play and Idle states, likely attempting to provide additional comfort to the robot after realizing its greater neediness compared to the robot in their first interaction. These results align with the behavior observed in the SP group, which displayed a higher frequency of touch during Interact and Recharge with the social robot.

Finally, participants frequently touched the robot in all states, including when it was recharging or sleeping, and adjusted the frequency of their touches based on the robot's profile. We also noticed high variability between participants, and the touch stimuli had a greater impact on the social robot's dynamics, leading to more variability between states during the interaction.

4) Robot's Stress Level Over Time: Finally, to understand the evolution of the robot's stress levels during interactions, we divided the interaction into three main phases: the beginning (INI, the first two minutes), the middle (MID, minutes 5 and 6), and the end (END, the last two minutes). These phases are illustrated in Fig. 10. As expected, the robot exhibited high-stress levels at the beginning of the interaction, likely due to the initial imbalance associated with the robot's profile drives and the participant's initial interactions exploring stimuli reactions and the robot's dynamics. As the interaction progressed, participants began understanding the robot and adjusting their behavior accordingly, reducing stress levels. The stress levels in the MID and END phases were similar, suggesting that participants maintained a consistent pattern of behavior, stabilizing the robot's drives.

We conducted a statistical analysis at two levels: (I) comparing the stress level of the same robot across three time windows (Beginning, Middle, End), and (II) comparing the stress level between the two robot profiles at each time window.

For the first level of analysis, we found significant differences only for the Playful robot in both groups. The Shapiro-Wilk test indicated that the data were distributed non-normally ($p < 0.050$). Subsequently, the Kruskal-Wallis test confirmed a significant decrease in stress levels across the three time windows ($p < 0.001$). The post-hoc Tukey test revealed significant differences between the beginning and middle of the interaction ($p < 0.001$), as well as between the beginning and end of the interaction ($p < 0.001$).

For the second level of analysis, the Shapiro-Wilk test again indicated a non-normal distribution of the data ($p < 0.050$), except for the beginning of the interaction (INI) in the SP group. We used the Mann-Whitney test for non-parametric data and the t-test for parametric data. Significant differences were found between the two robot profiles in all time windows for both groups: PS Group: Beginning (INI): $p < 0.001$, Middle (MID): $p = 0.003$, and End (END): $p < 0.001$; SP Group: Beginning (INI): $p < 0.001$, Middle (MID): $p = 0.030$, and End (END): $p = 0.018$. For both groups, the Playful profile exhibited higher

initial stress levels than the Social profile. However, this trend reversed as the interaction progressed, with the Social robot showing higher stress levels than the Playful robot during the middle and end of the interaction.

Comparing the groups, we observed that the PS group generally induced less stress in the robot. This can be attributed to the higher level of stimuli provided by the PS group, as shown in Fig. 8. The increased stimuli likely enhanced the robot's comfort (affect drive), reducing average stress levels. The playful profile is designed to be highly stressed at the beginning of the interaction. However, once participants master it, they can consistently make the robot happy, especially if it is the first profile they interact with. Conversely, keeping a socially needy robot satisfied is feasible from the beginning of the interaction but requires active participant involvement. The ability and willingness to engage actively vary significantly across participants.

Interestingly, those who first encountered the social profile exhibited higher robot stress levels during the subsequent playful session than the PS group. These results suggest that participants in the SP group noticed that the robot required fewer stimuli in the second session, leading to a quicker decrease in affection drive and, consequently, higher stress levels. In contrast, the PS group tended to interact more with the robot during their first interaction, unaware that the robot did not need as much stimulation, thereby increasing comfort and reducing average stress levels. Specifically for the playful robot, the higher stress levels showed by the SP group indicate that participants understood the robot's profile better by not providing unnecessary input stimuli, which increased stress faster.

V. Discussion

One of the most common objectives in the field of human-robot interaction is to enable robots to interact seamlessly with other agents, whether human or artificial. However, most robots are currently developed for specific contexts, limiting their applicability. Our hypothesis posits that if robots can develop similarly to humans, they could better learn and adapt based on their experiences. While cognitive robotics has significantly enhanced robots' learning, planning, and decision-making capabilities, it often lacks partner interactions. Conversely, human-robot interaction has achieved remarkable progress in social interaction, but the cognitive aspects of robots have not been deeply explored. Our work represents an initial step in integrating the strengths of both fields.

We developed a cognitive architecture that simulates certain human cognitive functions in a simplified manner, such as perception, motivation, and decision-making. These cognitive functions operate in parallel, with execution times varying according to their cognitive "level", with sensory processes being faster than decision-making ones. This non-blocking system allows the agent to continuously perceive the environment even during decision-making, facilitating the implementation of more complex behaviors and adaptation to rapid environmental changes. Our architecture is fully modular; although implemented in the iCub robot, it can be adapted for any robot by modifying the sensory and motor components to fit the agent's specific sensors and actuators.

Our autonomous agent acts based on motivations computed from internal states, influenced by environmental factors and the robot's needs. We applied Hull's drive reduction theory to compute the motivation levels for three primary needs: survival, affection, and exploration. These needs generate drives that indicate their urgency and guide the robot's behavior.

Social interaction necessitates understanding others and how interactions influence both parties. Therefore, we created a robot with intrinsic needs dependent on the partner to varying degrees—from complete independence to full dependency. The robot utilizes human partners to satisfy its needs, akin to a young child learning about itself and the impact of its actions and its partner's actions. Given the variability in human behavior and preferences, we designed two agents with the same cognitive architecture but different preferences to study human interactions with distinct agents. Our results indicate that participants adapted their behavior to help the robot achieve pleasure and well-being. For instance, when interacting with the social robot, participants provided more tactile and stimulus input, even when not requested, understanding the robot's higher neediness. Conversely, interactions with the playful profile differed, with participants engaging more with the toys and exploring their affordances, always showing new things to the robot. When the robot is idle, participants actively try to engage the robot's attention through physical touch, positioning, and vocal cues.

Our data analysis reveals distinct interaction patterns among the robot profiles, demonstrating participants' intuitive recognition of each robot's needs. This insight suggests potential for future research into collaborative tasks and the robot's understanding of its partner's needs, advancing toward mutual understanding based on the theory of mind concepts.

Participants frequently talked to the robots, although the robots lacked auditory functionality. This indicates the importance of equipping robots with the ability to detect speech, which could enhance their comfort levels. An interesting observation was that participants created new toys from available materials and mimicked real-life sounds, which could elicit surprise and curiosity from the robot. Enhancing the robot's ability to detect such changes would improve its learning process, leading to more natural and enjoyable interactions.

Predicting participant behavior in free-form interaction scenarios is challenging. Although we expected participants to respond to the robot's care requests through touch and eye contact, many spoke softly or played with toys to calm the robot down, which could increase the robot's comfort. Future work will explore better methods for perceiving various stimuli provided by humans.

Our current robot's needs function in an isolated manner and a change in one drive does not impact other elements in the architecture. However, in future work, we plan to integrate these needs into a more robust system inspired by humans, such that imbalances in one drive affect elements like cognitive functions (sensory and perceptual processes, for example) and other drives. For example, hunger could reduce interest in playing, or fatigue could slow decision-making. Additionally,

we will investigate the impacts of varying energy consumption associated with different behaviors.

While our architecture is inspired by human behavior, it currently lacks a mechanism for the robot to signal when it feels overwhelmed. Future work will address this to improve communication and enhance people's understanding of the robot's functionality, making interactions more similar to those between humans. Furthermore, it is essential for robots to understand their partner's needs and maintain a balance between autonomy, dependence, and assisting others.

Affective computing principles offer a compelling framework for achieving this balance, as they emphasize equipping robots with the ability to process and simulate human emotions. Robots driven by survival, exploration, and affective motivations embody these principles by combining functional autonomy with emotional responsiveness. The affective drive, in particular, enables robots to interpret and respond to social cues, fostering more natural and engaging interactions. This integration allows robots to adapt their behaviors in real-time, enhancing their capacity to meet both their own needs and those of their human partners in a harmonious manner.

The affective drive significantly impacted the robot's behavior and participant perception during experiments. By prioritizing actions based on human social cues, such as touch or facial expression, the robot appeared more attentive and relatable. Participants naturally responded to this by increasing their engagement, providing tactile feedback, and interacting with objects to sustain the robot's interest. This interplay highlighted how affective drives can elicit and maintain meaningful social responses, making the robot a more effective partner in human-robot interaction scenarios.

The integration of affective motivation demonstrates the potential of emotionally intelligent robots to function as adaptive and engaging agents. Such systems not only enhance user satisfaction and trust but also underscore the importance of combining emotional responsiveness with functional autonomy. Future developments may focus on expanding the affective module to handle more nuanced cues and employ empathy considering the partner's expressed emotion, further enhancing the robot's ability to adapt to diverse human emotional dynamics and fostering deeper human-robot collaboration.

A. *Limitations*

This study presents valuable insights into the design of curiosity and affect-driven cognitive architecture for robots, though it also highlights several limitations that offer important directions for future research. The controlled experimental setup was designed to reduce complexity and address experimental challenges. It involved only simple interactions within a close workspace and with one human at a time. The types of objects and robot actions were constrained, and the range of recognized sensory stimuli was limited. However, the architecture itself is flexible, allowing for the addition of new drives, actions, and perceptual capabilities. These modifications allow the robot to adapt dynamically, laying the groundwork for future studies on more complex and generalizable robotic systems for real-world applications. We envision the generalization of our system to more open-ended scenarios, such as, for instance, art installations where the effectiveness of a similar system has been previously demonstrated [30], [31], [32]). Furthermore, the ability of drives in our architecture to modulate the attention system adds a layer of sophistication, enabling the system to support more diverse and dynamic interactions across varied contexts.

The robot's operation was based on three basic needs, chosen to simplify the exploration of fundamental behaviors and their influence on decision-making and interaction. While this simplification was necessary for understanding human perceptions of a cognitive autonomous agent, it limits the scope of the study. Future research could explore methods for creating new drives autonomously or refining existing ones by subdividing them into more nuanced categories. The challenge of autonomously generating drives warrants further investigation. An exciting direction involves enabling the robot to transfer its learning across scenarios, potentially through reinforcement learning algorithms, which would allow the robot to better understand its needs and adapt to human behavior without requiring predefined contexts.

Another limitation is the robot's lack of adaptive mechanisms for interacting with different agents. Future work should prioritize equipping the robot with learning capabilities to generalize its behavior and continuously adapt to diverse environments [71] and individuals [72]. While this poses significant challenges, it is essential for creating autonomous systems capable of meaningful interaction. Data from this study will help in developing algorithms that allow the robot to interpret its needs, understand partner responses, and select effective actions. Additionally, future research could integrate social cognition elements, such as the "Like Me" theory [73], [74], to enhance the robot's ability to assist human partners.

Building on this foundation, subsequent studies have extended the architecture by integrating Theory of Mind components, enhancing the robot's capacity to understand and engage in collaborative decision-making with humans and other robots. These improvements are crucial for fostering harmonious interactions in social and collaborative environments.

A key limitation is the participant-specific nature of the findings and due to the relatively small sample size of the experiment. To generalize these results, future experiments should involve a larger, more diverse group of participants across varied locations and cultural backgrounds. This would provide a broader understanding of how these factors influence human-robot interaction and perceptions. Additionally, exploring prolonged interactions with the robot over extended periods or repeated sessions could reveal how people's perceptions evolve. While the robot in this study has memory capabilities, these were not leveraged for interaction enhancement. Future work should incorporate memory to improve interaction by recalling past experiences and informing future decisions.

VI. Conclusion

In conclusion, using two distinct profiles, we investigated how people interact with them without being aware of their

differences, similar to human interactions with strangers. Our findings suggest that simply switching priorities among drives can result in robots behaving and being perceived differently by human partners. Although the robot has its preferences, the interaction dynamics depend significantly on partner engagement. In our experiments, iCub interacted with many different people, creating variability in the interactions. Despite this variability, the robot's behavior was compatible with the robot profiles, and the robot was understood by most participants, especially when exhibiting playful behavior. This underscores the importance of a robot's "personality," as it influences how people perceive and interact [75], [76].

Although the robot currently relies on human assistance to meet its needs, future developments will aim for the robot to learn about itself, understand human stimuli and internal dynamics, adapt by learning from past interactions, allowing it to adjust its behavior accordingly, and eventually recognize and help satisfy human needs. Another key aspect is balancing the robot's autonomy with its neediness, ensuring it remains neither too detached nor too needy. Striking a balance between autonomy and neediness is crucial for robots to engage effectively with humans. A too detached robot may seem unresponsive, while one that is too needy can be overwhelming. Balanced robots can independently pursue goals while seeking human help when needed, adapting to different contexts and user preferences. This balance is essential in social robotics, where the primary role is to interact meaningfully with humans, as being too detached undermines this role, and being too needy disrupts social harmony.

Addressing the limitations and expanding the framework will enable future research to develop autonomous, adaptive robots capable of complex interactions in diverse settings. The robot profiles used in this study, designed for human-robot interaction and cognitive exploration, also have potential applications across multiple domains. These include healthcare, education, customer service, personal companionship, rehabilitation, workplace collaboration, and smart city initiatives. The modular nature of the architecture allows it to be easily adapted to these contexts, providing scalability without the need for complete redesigns.

Acknowledgment

The authors would like to thank Sara Mongile, Francesca Cocchella, and Marco Fedozzi for their help in designing the experiment and the questionnaires.

References

[1] B. Lindblom, "Explaining phonetic variation: A sketch of the H&H theory," in *Speech Production and Speech Modelling*. Berlin, Germany: Springer, 1990, pp. 403–439.

[2] A. Savidis and C. Stephanidis, "Unified design for user interface adaptation," in *The Universal Access Handbook*, C. Stephanidis Ed. New York, NY, USA: Taylor & Francis, 2009, ch. 16, pp. 16-1–16-17.

[3] A. Mehrabian and N. Epstein, "A measure of emotional empathy 1," *J. Pers.*, vol. 40, no. 4, pp. 525–543, 1972.

[4] H. Nakayama, "Emergence of amae crying in early infancy as a possible social communication tool between infants and mothers," *Infant Behav. Develop.*, vol. 40, pp. 122–130, 2015.

[5] P. Dunstan, *Calm the Crying: The Secret Baby Language That Reveals the Hidden Meaning Behind an Infant's Cry*. Baltimore, MD, USA: Penguin, 2012.

[6] M. Chóliz, E. G. Fernández-Abascal, and F. Martínez-Sánchez, "Infant crying: Pattern of weeping, recognition of emotion and affective reactions in observers," *Spanish J. Psychol.*, vol. 15, no. 3, pp. 978–988, 2012.

[7] M. Tomasello, *Constructing a Language: A Usage-Based Theory of Language Acquisition*. Cambridge, MA, USA: Harvard Univ. Press, 2005.

[8] L. J. Wood, A. Zaraki, M. L. Walters, O. Novanda, B. Robins, and K. Dautenhahn, "The iterative development of the humanoid robot Kaspar: An assistive robot for children with Autism," in *Proc. Int. Conf. Social Robot.*, Tsukuba, Japan, Springer, 2017, pp. 53–63.

[9] C. Plaisant et al., "A storytelling robot for pediatric rehabilitation," in *Proc. 4th Int. ACM Conf. Assistive Technol.*, Arlington, USA, 2000, pp. 50–55.

[10] H. Admoni and B. Scassellati, "Data-driven model of nonverbal behavior for socially assistive human-robot interactions," in *Proc. 16th Int. Conf. Multimodal Interact.*, Istanbul, Turkey, 2014, pp. 196–199.

[11] C. D. Kidd, W. Taggart, and S. Turkle, "A sociable robot to encourage social interaction among the elderly," in *Proc. 2006 IEEE Int. Conf. Robot. Automat.*, 2006, pp. 3972–3976.

[12] E. Broadbent, C. Jayawardena, N. Kerse, R. Q. Stafford, and B. A. MacDonald, "Human-robot interaction research to improve quality of life in elder care—An approach and issues," in *Proc. 25th AAAI Conf. Artif. Intell. Workshops*, San Francisco, USA, 2011, pp. 13–19.

[13] A. Sharkey, "Robots and human dignity: A consideration of the effects of robot care on the dignity of older people," *Ethics Inf. Technol.*, vol. 16, no. 1, pp. 63–75, 2014.

[14] A. Ramachandran, A. Litoiu, and B. Scassellati, "Shaping productive help-seeking behavior during robot-child tutoring interactions," in *Proc. 11th ACM/IEEE Int. Conf. Hum. Robot Interact.*, 2016, pp. 247–254.

[15] F. Jimenez, T. Yoshikawa, T. Furuhashi, and M. Kanoh, "An emotional expression model for educational-support robots," *J. Artif. Intell. Soft Comput. Res.*, vol. 5, no. 1, pp. 51–57, 2015.

[16] A. Paiva, I. Leite, and T. Ribeiro, "Emotion modeling for social robots," in *The Oxford Handbook of Affective Computing*, R. Calvo, S. D'Mello, J. Gratch, and A. Kappas, London, U.K.: Oxford Univ. Press, 2014, pp. 296–308.

[17] F. Tanaka and S. Matsuzoe, "Children teach a care-receiving robot to promote their learning: Field experiments in a classroom for vocabulary learning," *J. Hum.-Robot Interact.*, vol. 1, no. 1, pp. 78–95, 2012.

[18] M. I. Ahmad, O. Mubin, S. Shahid, and J. Orlando, "Robot's adaptive emotional feedback sustains children's social engagement and promotes their vocabulary learning: A long-term child–robot interaction study," *Adaptive Behav.*, vol. 27, no. 4, pp. 243–266, 2019.

[19] D. Vaufreydaz, W. Johal, and C. Combe, "Starting engagement detection towards a companion robot using multimodal features," *Robot. Auton. Syst.*, vol. 75, pp. 4–16, 2016.

[20] C. Breazeal and B. Scassellati, "How to build robots that make friends and influence people," in *Proc. 1999 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 1999, pp. 858–863.

[21] L. Cañamero, A. J. Blanchard, and J. Nadel, "Attachment bonds for human-like robots," *Int. J. Humanoid Robot.*, vol. 3, no. 03, pp. 301–320, 2006.

[22] T. Kishi et al., "Bipedal humanoid robot that makes humans laugh with use of the method of comedy and affects their psychological state actively," in *Proc. 2014 IEEE Int. Conf. Robot. Automat.*, Hong Kong, China, 2014, pp. 1965–1970.

[23] A. Cangelosi et al., "Integration of action and language knowledge: A roadmap for developmental robotics," *IEEE Trans. Auton. Ment. Develop.*, vol. 2, no. 3, pp. 167–195, Sep. 2010.

[24] D. Vernon, *Artificial Cognitive Systems: A Primer*. Cambridge, MA, USA: MIT Press, 2014.

[25] D. McColl, A. Hong, N. Hatakeyama, G. Nejat, and B. Benhabib, "A survey of autonomous human affect detection methods for social robots engaged in natural HRI," *J. Intell. Robot. Syst.*, vol. 82, no. 1, pp. 101–133, 2016.

[26] C. Adam, W. Johal, D. Pellier, H. Fiorino, and S. Pesty, "Social human-robot interaction: A new cognitive and affective interaction-oriented architecture," in *Proc. Int. Conf. Social Robot.*, 2016, pp. 253–263.

[27] A. Tanevska, F. Rea, G. Sandini, L. Cañamero, and A. Sciutti, "A socially adaptable framework for human-robot interaction," *Front. Robot. AI*, vol. 7, 2020, Art. no. 121. [Online]. Available: https://www.frontiersin.org/articles/10.3389/frobt.2020.00121

[28] S. Lemaignan, M. Warnier, E. A. Sisbot, A. Clodic, and R. Alami, "Artificial cognition for social human–robot interaction: An implementation," *Artif. Intell.*, vol. 247, pp. 45–69, 2017.

[29] J. Beer, A. D. Fisk, and W. A. Rogers, "Toward a framework for levels of robot autonomy in human-robot interaction," *J. Hum. Robot Interact.*, vol. 3, no. 2, 2014, Art. no. 74.
[30] C. Kroos, D. C. Herath, and S. Stelarc, "Evoking agency: Attention model and behavior control in a robotic art installation," *Leonardo*, vol. 45, no. 5, pp. 401–407, 2012.
[31] C. Kroos and D. C. Herath, "We, robots: Correlated behaviour as observed by humans," in *Proc. 6th Int. Conf. Social Robot.*, Springer, 2014, pp. 229–238.
[32] L. Cavedon et al., ""C'Mon dude!": Users adapt their behaviour to a robotic agent with an attention model," *Int. J. Hum. Comput. Stud.*, vol. 80, pp. 14–23, 2015.
[33] J. E. Laird, *The Soar Cognitive Architecture*. Cambridge, MA, USA: MIT Press, 2012.
[34] J. R. Anderson, D. Bothell, M. D. Byrne, S. Douglass, C. Lebiere, and Y. Qin, "An integrated theory of the mind," *Psychol. Rev.*, vol. 111, no. 4, 2004, Art. no. 1036.
[35] P. S. Rosenbloom, A. Demski, and V. Ustun, "The sigma cognitive architecture and system: Towards functionally elegant grand unification," *J. Artif. Gen. Intell.*, vol. 7, no. 1, pp. 1–103, 2016.
[36] R. Sun, "The CLARION cognitive architecture: Extending cognitive modeling to social simulation," in *Cognition and Multi-Agent Interaction*. Cambridge, U.K.: Cambridge Univ. Press, 2006, pp. 79–99.
[37] D. Choi and P. Langley, "Evolution of the ICARUS cognitive architecture," *Cogn. Syst. Res.*, vol. 48, pp. 25–38, 2018.
[38] D. Vernon, G. Metta, and G. Sandini, "The iCub cognitive architecture: Interactive development in a humanoid robot," in *Proc. IEEE 6th Int. Conf. Develop. Learn.*, 2007, pp. 122–127.
[39] I. Kotseruba and J. K. Tsotsos, "40 years of cognitive architectures: Core cognitive abilities and practical applications," *Artif. Intell. Rev.*, vol. 53, no. 1, pp. 17–94, 2020.
[40] P. Langley, J. E. Laird, and S. Rogers, "Cognitive architectures: Research issues and challenges," *Cogn. Syst. Res.*, vol. 10, no. 2, pp. 141–160, 2009.
[41] K. Thórisson and H. Helgasson, "Cognitive architectures and autonomy: A comparative review," *J. Artif. Gen. Intell.*, vol. 3, no. 2, pp. 1–30, 2012.
[42] C. Watkins, "Learning from delayed rewards," PhD thesis, Kings College, Cambridge Univ., Cambridge, U.K., 1989.
[43] A. H. Maslow, *Motivation and Personality*. New Delhi, India: Prabhat Prakashan, 1981.
[44] A. H. Maslow, "A theory of human motivation," *Psychol. Rev.*, vol. 50, no. 4, 1943, Art. no. 370.
[45] G. Metta, G. Sandini, D. Vernon, L. Natale, and F. Nori, "The iCub humanoid robot: An open platform for research in embodied cognition," in *Proc. 8th Workshop Perform. Metrics Intell. Syst.*, 2008, pp. 50–56.
[46] G. Metta et al., "The iCub humanoid robot: An open-systems platform for research in cognitive development," *Neural Netw.*, vol. 23, no. 8, pp. 1125–1134, 2010. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608010001619
[47] A. Parmiggiani et al., "The design of the iCub humanoid robot," *Int. J. Humanoid Robot.*, vol. 9, no. 04, 2012, Art. no. 1250027.
[48] G. Sandini, G. Metta, and D. Vernon, "The iCub cognitive humanoid robot: An open-system research platform for enactive cognition," in *50 Years of Artificial Intelligence*. Berlin, Germany: Springer, 2007, pp. 358–369.
[49] R. Feldman, "Infant–mother and infant–father synchrony: The coregulation of positive arousal," *Infant Ment. Health J. Official Pub. World Assoc. Infant Ment. Health*, vol. 24, no. 1, pp. 1–23, 2003.
[50] P. Ekman and W. V. Friesen, *Facial Action Coding System: A Technique for the Measurement of Facial Movement*, Palo Alto, CA: Consulting Psychologists Press, 1978.
[51] T. Baltrušaitis, P. Robinson, and L.-P. Morency, "OpenFace: An open source facial behavior analysis toolkit," in *Proc. 2016 IEEE Winter Conf. Appl. Comput. Vis.*, 2016, pp. 1–10.
[52] G. Cannata, M. Maggiali, G. Metta, and G. Sandini, "An embedded artificial skin for humanoid robots," in *Proc. 2008 IEEE Int. Conf. Multisensor Fusion Integration Intell. Syst.*, 2008, pp. 434–438.
[53] S. Mongile, A. Tanevska, F. Rea, and A. Sciutti, "Validating a cortisol-inspired framework for human-robot interaction with a replication of the still face paradigm," in *Proc. 2022 IEEE Int. Conf. Develop. Learn.*, 2022, pp. 190–196.
[54] O. Avila-Garcia and L. Canamero, "Hormonal modulation of perception in motivation-based action selection architectures," in *Proc. Symp. Agents Want Like*, 2005, pp. 9–16.
[55] A. Hiolle, L. Cañamero, M. Davila-Ross, and K. A. Bard, "Eliciting caregiving behavior in dyadic human-robot attachment-like interactions," *ACM Trans. Interactive Intell. Syst.*, vol. 2, no. 1, 2012, Art. no. 3.
[56] M. D. S. Ainsworth, M. C. Blehar, E. Waters, and S. N. Wall, *Patterns of Attachment: A Psychological Study of the Strange Situation*. London, U.K.: Psychology Press, 2015.
[57] R. Feldman, M. Singer, and O. Zagoory, "Touch attenuates infants' physiological reactivity to stress," *Devlop. Sci.*, vol. 13, no. 2, pp. 271–278, 2010.
[58] W. B. Cannon, *The Wisdom of the Body*. New York, NY, USA: Norton, 1939.
[59] A. Clark, *Being There: Putting Brain, Body, and World Together Again*. Cambridge, MA, USA: MIT Press, 1998.
[60] F. J. Varela, E. Thompson, and E. Rosch, *The Embodied Mind: Cognitive Science and Human Experience*, Revised ed. Cambridge, MA, USA: MIT Press, 2017.
[61] M. Merleau-Ponty, "Phenomenology of perception," *Humanities*, D. Landes, Trans.; 1st ed. Routledge, 1962, doi: 10.4324/9780203720714.
[62] S. D. Gosling, P. J. Rentfrow, and W. B. Swann Jr, "A very brief measure of the big-five personality domains," *J. Res. Pers.*, vol. 37, no. 6, pp. 504–528, 2003.
[63] F. Ferrari, M. P. Paladino, and J. Jetten, "Blurring human–machine distinctions: Anthropomorphic appearance in social robots as a threat to human distinctiveness," *Int. J. Social Robot.*, vol. 8, pp. 287–302, 2016.
[64] F. Spaccatini, M. G. Pacilli, I. Giovannelli, M. Roccato, and G. Penone, "Sexualized victims of stranger harassment and victim blaming: The moderating role of right-wing authoritarianism," *Sexuality Culture*, vol. 23, no. 3, pp. 811–825, 2019.
[65] G. Pusceddu et al., "Training school teachers to use robots as an educational tool: The impact on robotics perception," in *Proc. Int. Conf. Social Robot.*, Springer, 2022, pp. 103–113.
[66] S. T. Fiske, A. J. Cuddy, and P. Glick, "Universal dimensions of social cognition: Warmth and competence," *Trends Cogn. Sci.*, vol. 11, no. 2, pp. 77–83, 2007.
[67] H. M. Gray, K. Gray, and D. M. Wegner, "Dimensions of mind perception," *Science*, vol. 315, no. 5812, pp. 619–619, 2007.
[68] S. Gächter, C. Starmer, and F. Tufano, "Measuring the closeness of relationships: A comprehensive evaluation of the 'inclusion of the other in the self' scale," *PLoS One*, vol. 10, no. 6, 2015, Art. no. e0129478.
[69] L. J. Cronbach, "Coefficient alpha and the internal structure of tests," *Psychometrika*, vol. 16, no. 3, pp. 297–334, 1951.
[70] C. Chiorri, *Psicometria e teoria dei test*. McGraw Hill Create, 2016, Art. no. 748.
[71] L. Berto, P. Costa, A. Simões, R. Gudwin, and E. Colombini, "A motivational-based learning model for mobile robots," *Cogn. Syst. Res.*, vol. 88, 2024, Art. no. 101278. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S138904172400072X
[72] M. Liu and C. Chen, "Task decoupling in preference-based reinforcement learning for personalized human-robot interaction," in *Proc. 2022 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2022, pp. 848–855.
[73] A. N. Meltzoff, "The 'like me' framework for recognizing and becoming an intentional agent," *Acta Psychologica*, vol. 124, no. 1, pp. 26–43, 2007.
[74] A. N. Meltzoff, "'Like me': A foundation for social cognition," *Devlop. Sci.*, vol. 10, no. 1, pp. 126–134, 2007.
[75] M. Paetzel-Prüsmann, G. Perugia, and G. Castellano, "The influence of robot personality on the development of uncanny feelings," *Comput. Hum. Behav.*, vol. 120, 2021, Art. no. 106756.
[76] A. Andriella et al., "Do I have a personality? Endowing care robots with context-dependent personality traits," *Int. J. Social Robot.*, vol. 13, pp. 2081–2102, 2021.



Letícia Berto is currently working toward the PhD degree with the State University of Campinas. Her research is grounded in developing cognition in robots, drawing inspiration from child development. Her primary focus is constructing cognitive autonomous robots by implementing cognitive architectures, emphasizing intrinsic motivation, learning, and decision-making processes. Her research interests include human-robot interaction, affective computing, cognitive science, artificial intelligence, and theory of mind.



Ana Tanevska is a postdoctoral researcher within the Horizon Europe Project SymAware. Their research at the Social Robotics Lab, Uppsala University is grounded in the topic of trustworthy human-robot interaction (HRI), with a particular focus on human-in-the-loop learning, and shared autonomy and agency in HRI. In addition to trustworthy HRI, her research interests include cognitive robotics (with a spotlight on the role of affect in cognition), and socially-assistive HRI.



Azamor Cirne received the BS degrees in mining engineering and in civil engineering from the Federal University of Paraíba (UFPB), the MS degree in mechanical engineering from the Federal University of Paraíba, and the PhD degree in mechanical engineering from the State University of Campinas (UNICAMP), Brazil. He is an associate professor with the Department of Finance and Accounting, Center for Applied Social Sciences, Federal University of Paraíba. He works in quantitative methods, statistics, and data analysis.



Paula Costa received the BS degree in electrical engineering from UNICAMP, Brazil, the MS degree in mobile communication from the Polytechnique of Torino, Italy, and the MS and PhD degrees in computer engineering from UNICAMP, Brazil. She is a professor with the Faculty of Electrical and Computer Engineering, State University of Campinas (UNICAMP), Brazil. Her research interests include multimodal artificial intelligence, affective computing, and science disclosure for young people.



Alexandre Simões received the BS degree in electrical engineering from the São Paulo State University (UNESP), Brazil, and the MS and PhD degrees in electrical engineering from the Polytechnic School, São Paulo University, Brazil. He is an associate professor with the Institute of Science and Technology, Campus Sorocaba, São Paulo State University (Unesp), Brazil. His research interests include robotics, machine learning, cognitive systems, and the interface between these research fields, science disclosure, education, and arts.



Ricardo Gudwin received the BS, MS, and PhD degrees in electrical engineering from the Computação, Universidade Estadual de Campinas (UNICAMP), Brazil. He is an associate professor with the Faculdade de Engenharia Elétrica e de Computação, Universidade Estadual de Campinas, Brazil. He is the head of the "Computational Semiotics Group", and the "Group for Research on Artificial Cognition" and a co-PI of CEPID BRAINN. His research interests include cognitive architectures and intelligent systems and agents.



Francesco Rea received the bachelor's degree in software engineering from the Universita di Bergamo Italy, the MSc (with distinction) degree in robotics and automation from Salford University, England, and the PhD degree, in 2013. Since 2009, with the Istituto di Tecnologia, he has promoted the research in the field of humanoid robotics. He is now team leader with the Cognitive Interaction Lab and principal investigator with IIT for related European Projects.



Esther Colombini received the MSc and PhD degrees in computing engineering from the Technological Institute of Aeronautics, Brazil. She is a professor with the University of Campinas, Brazil, where she coordinates the Laboratory of Robotics and Cognitive Systems. Her research interests include machine learning and AI for autonomous robotics and healthcare, focusing on reinforcement learning, attentive models, and cognitive modeling.



Alessandra Sciutti received the BS and MS degrees in bioengineering and the PhD degree in humanoid technologies from the University of Genova, in 2010. She is tenure track researcher, head with the CONTACT (COgNiTive Architecture for Collaborative Technologies) Unit, Italian Institute of Technology. She has been awarded the ERC Starting Grant wHiSPER. Her research aims to investigate the sensory and motor mechanisms underlying mutual understanding in human-human and human-robot interaction.

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - ROR identifier: 00x0ma614